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Differences in time–frequency representation of lower limbs myoelectric activity during single and double leg landing in male athletes

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ABSTRACT

This study compared the instantaneous median frequency (IMF) obtained by means of a Choi-Williams transform of an electromyogram of the lower-limb muscles during single-leg (SL) and double-leg (DL) landings performed by fifteen male athletes. The IMF values of the rectus femoris (RF), biceps femoris (BF) and hip adductors (HA) were compared between two landing tasks, within each landing, and before and after ground contact (GC). The IMF values of the RF did not change between landings in contrast to those of the BF, which presented from 20- to 40-ms higher SL values before GC and from 40 to 60 ms after GC. HA presented higher SL values during the 40–60 ms range before GC. Within each landing, the RF IMF decreased from 40 ms to 60 ms after GC in the SL. Similar results were found for the HA IMF, which decreased from 40 ms to 80 ms after GC. The BF IMF showed no significant change. These results suggest muscle recruitment related to anterior cruciate ligament protection since the IMF values of the RF decreased in the SL, whereas the BF IMF increased. Results for the HA showed the importance of hip muscles in stabilizing the core region, allowing the activation of distal muscles with greater safety.

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1. Introduction

Although much attention has been devoted to discovering the causal and predictive factors related to anterior cruciate ligament (ACL) injury, the available data support few conclusive results (Alentorn-Geli et al., 2009). A variety of evidence associates anatomical, hormonal and environmental factors to this lesion (Griffin et al., 2006; Renstrom et al., 2008). However, the neuromuscular and biomechanical characteristics are primarily studied because many variables can be modified by an adjusted training strategy (Renstrom et al., 2008). This approach has shown great effectiveness in reducing the incidence of injuries (Grindstaff et al., 2006).

Among the biomechanical variables, the knee valgus and the flexion angle during landing tasks, which are both associated with vertical ground reaction forces, present a greater causal relationship with ACL injuries, evidenced by prospective epidemiological (Hewett et al., 2005a), in vitro (Withrow et al., 2006) and simulation (McLean et al., 2008; Shin et al., 2009) studies.

Neuromuscular variables have assumed an important role in recent years. Although studies have already demonstrated that excessive activation of the quadriceps muscle, in relation to the hamstrings, increases tension in the ACL in controlled laboratory conditions (Draganich and Vahey, 1990; Li et al., 1999; DeMorat et al., 2004), only recently has a study shown a direct relationship between quadriceps/hamstring activation rates and ACL injuries (Zebis et al., 2009).

Another neuromuscular characteristic that has been extensively investigated is the relationship between the lumbar-pelvic region control, also called core region, and ACL injuries. Leetun et al. (2004) showed a strong correlation between trunk and hip muscle strength and knee injuries. Zazulak et al. (2007) reported that factors related to core stability, such as trunk displacement and proprioception, predicted ACL injuries in women, with 91% sensitivity and 68% specificity.

Recently, new strategies for myoelectric signal processing related to the time–frequency representation (TFR) have been applied, revealing valuable information and important interpretations concerning the motor units (MU) recruitment (Von Tscharner and Goepfert, 2003; Wakeling et al., 2002).

Several techniques have been developed to analyze changes in the signal spectral characteristics over time, among them is the exponential distribution proposed by Choi and Williams (1989), which belongs to the Cohen class of transforms (Cohen, 1989). The use of this class of transforms allows the estimation of new parameters, such as the instantaneous median frequency (IMF)

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and the instantaneous power (Choi and Williams, 1989). The IMF corresponds to the frequency value that divides the instantaneous power into two equal parts at each time instant. Based on these definitions, it was proposed to use the IMF, as an alternative to the Fourier median frequency, to monitor the Choi–Williams distribution's (CWD) spectral content (Roy et al., 1998; Molinari et al., 2006).

While some authors suggest that the instantaneous mean and median frequencies would be related to the conduction velocity of the electrical potential in the muscle fiber and, consequently, to the muscle fiber type (Wakeling et al., 2002; White et al., 2003; Wakeling and Rozitis, 2004), other authors affirm that this relationship is not observed in dynamic situations (Farina et al., 2002; Merlo et al., 2005). However, recent studies with more controlled experimental models have confirmed the relationship between the frequency of activation, measured by the TFR of EMG signals, and the predominance of active MU during muscle contraction (Wakeling, 2009).

Of all the studies that discussed the MU recruitment by means of TFR, only one has examined motor tasks similar to those related to ACL injuries. Beaulieu et al. (2008) showed that during running, with unanticipated changes of direction, women showed lower values than men for the instantaneous mean frequency in hamstrings muscles. Similar results were presented by von Tscharner and Goepfert (2003) and White et al. (2003) during running tasks and knee flexion/extension on an isokinetic dynamometer, respectively. These authors concluded that these muscle recruitment strategies reduce the efficiency of the muscles around the knee in generating co-contraction and, consequently, stabilize the joint, pre-disposing women to ACL injuries.

No study was found to have examined muscle coordination related to different mechanical constraints, such as single-leg (SL) and double-leg (DL) landings. When measured in individuals with low-ACL injury risk, this information becomes valuable for developing different but effective and safe strategies for muscle recruitment, providing a reference basis for comparison with other populations with increased risk, such as female athletes.

The purpose of this study was to compare the IMF, obtained from the CWD of lower limb muscle EMG signals, between SL and DL landings in males. It was hypothesized that athletes would present different muscle recruitment strategies between landings with higher IMF values for hamstrings and HA and lower values for the rectus femoris during SL compared to DL. Because SL landings require the entire weight of the body to be supported by only one lower limb, an increased need for joint stabilization is required, condition that is decreased during DL landings.

2. Materials and methods

Fifteen male athletes from a regional volleyball team (age, 13 ± 1 year, height, 1.70 ± 0.12 m and weight, 60 ± 12 kg), with at least three years of experience playing volleyball, participated in this study. The athletes had no history of ligament injuries, nor had they reported pain at the time of the tests. All the players' parents signed an informed consent form allowing participation in the study. This study was approved by the State University of Rio de Janeiro Research Ethics Council.

Each subject performed two types of vertical jumps using both legs for propulsion and SL or DL landing (Fig. 1). The athletes performed the actions initially to familiarize themselves with the studied motor tasks. Thereafter, each athlete performed three landings with the dominant leg and three with both legs. Dominance was determined by questioning the athletes about their preferred leg for kicking a ball. The order of the jumps was randomized to minimize the possible effects of fatigue or learning. A one minute interval was allowed among trials. The EMG was captured from muscles of the dominant leg.

Ag/AgCl surface KOBME electrodes (Bio Protection, Korea) were positioned on the rectus femoris (RF), biceps femoris (BF) and hip adductors (HA) according to Cram et al. (1998). HA electrodes refer to the activity of adductor longus and gracilis muscles. RF electrodes were placed 2 cm apart, parallel to muscle fibers on the center of the anterior surface of the thigh, at approximately half the distance between the knee and the anterior superior iliac spine. BF electrodes were placed 2 cm apart, parallel to muscle fibers in the lateral thigh surface, approximately two-thirds the distance between the greater trochanter and the posterior knee joint. Finally, HA electrodes were positioned parallel to the muscle fibers in an oblique direction, 2 cm apart on the medial aspect of the thigh, approximately 4 cm from the pubis.

Before the application of the electrodes, the skin was prepared by shaving the area and cleansing with alcohol to reduce surface impedance. To prevent movement artifacts in the signals, the electrode cables were fixed to the skin using adhesive tape (3M, Brazil).

The EMG of all muscles were captured (EMG 100B, BIOPAC Systems, USA), amplified (differential bipolar amplification, input impedance = 2 MΩ, CMRR > 110 dB, gain = 1000), converted from analog to digital (12 bit, 2 kHz, MP100WSW BIOPAC Systems, USA) and stored on a personal computer.

To determine the initial ground contact, an electrical circuit was structured so that a terminal located in the sole in the first metatarsus region of the subject's shoe emitted a digital electrical signal upon contact with a metal platform fixed in the ground that was then captured by the UMI 100B module (BIOPAC Systems, USA).

The EMG signals were filtered by a 4th-order Butterworth filter (20–400 Hz), applied in direct and reverse directions to avoid phase distortions. After that, the CWD was obtained from 100-ms before initial ground contact (BGC) to 100-ms after initial ground contact (AGC). Mean IMF values were thus calculated for each 20-ms interval of the selected signal, totaling ten values (Table 1). The discrete CWD is given by:

$$CW(n,\theta) = 2\sum_{\tau=-\infty}^{\infty} \omega_n(\tau) e^{-j\theta\tau} \left(\sum_{\mu=-\infty}^{\infty} \omega_m(\mu) \frac{1}{\sqrt{\frac{4\pi\tau^2}{\sigma}}} e^{\frac{-\mu^2}{\frac{4\tau^2}{\sigma}}} (R(t,\tau)) \right) \quad (1)$$

where $\omega_n(\tau)$ is a symmetric window that presents non zero values in the $-N/2 \leqslant \tau \leqslant N/2$ interval, $\omega_m(\mu)$ is a Hamming window along the $-M/2 \leqslant \mu \leqslant M/2$ interval, R is the instantaneous autocorrelation function, represented by $x(n+\mu+\tau)\, x^*(n+\mu-\tau)$, and σ is a scaling factor ($\sigma = 1$). MATLAB version 6.5 (The Mathworks, USA) was used for signal processing.

As IMF values did not show adherence to a Gaussian distribution (Kolmogorov–Smirnov test), the paired mean IMF values between SL and DL were compared using the Wilcoxon Ranked Test. The Friedman's analysis of variance test with the post hoc test of Dunn was used to compare the IMF behavior between each 20-ms interval within each landing test.

The significance level determined for this study was at 5%. All statistical analyses were performed using the software Statistical Package for Social Sciences for Windows (version 13.0, SPSS, Chicago, IL).

3. Results

3.1. Comparison of IMF values between SL and DL landings

The RF showed no significant differences in the IMF values between the landings in any situation (Table 2 and Fig. 2a). The BF showed significant differences between the landings in two intervals: BGC2 and AGC3. The values of IMF were higher for the SL

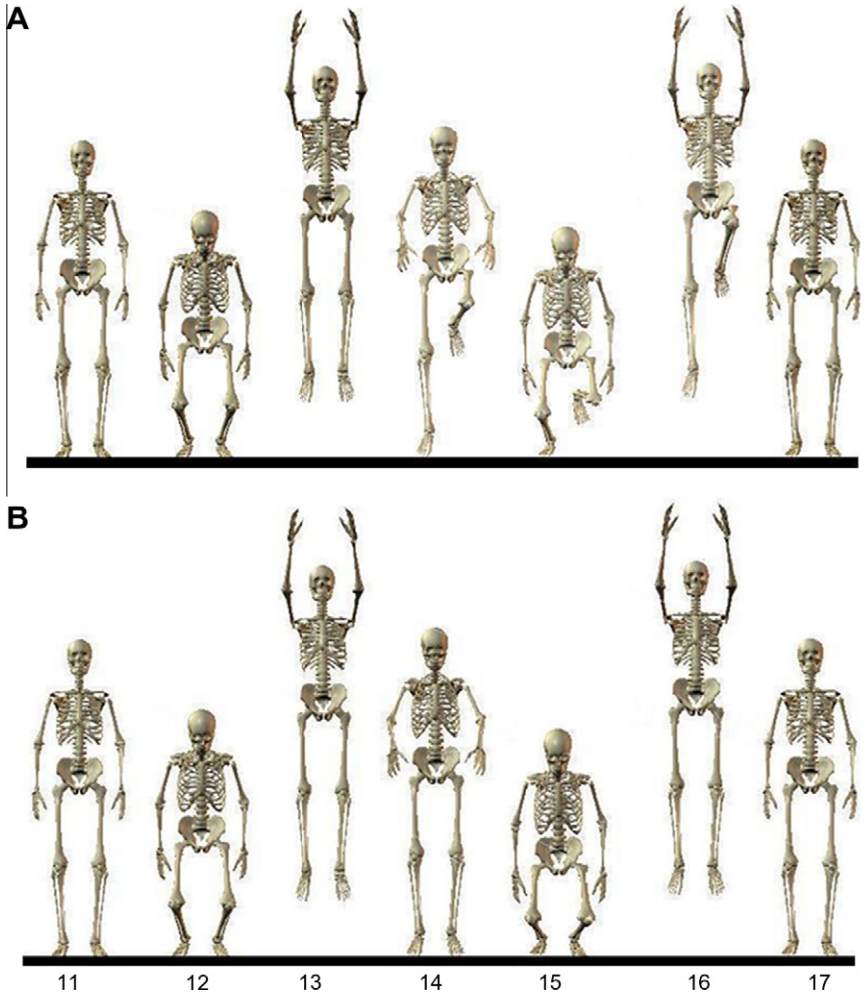

Fig. 1. Landing tasks used in this study. I4 Represents initial ground contact. (A) Single leg landing and (B) double leg landing.

Table 1
Periods of mean IMF value computation.

Before ground contact (BGC)		After initial ground contact (AGC)	
Period (ms)	Nomenclature	Period (ms)	Nomenclature
100–80	BGC5	0–20	AGC1
80–60	BGC4	20–40	AGC2
60–40	BGC3	40–60	AGC3
40–20	BGC2	60–80	AGC4
20–0	BGC1	80–100	AGC5

Table 2
Instantaneous median frequency (IMF) of rectus femoris in each 20 ms interval. The results are expressed as mean (standard error). The *p* value described in the forth column results from a comparison of between landing tasks, while the *p* values in the last line result from within landing comparisons.

	Single leg	Double leg	*p* Value
BGC5	141.9(4.6)	147.6(7.4)	0.4631
BGC4	148.8(5.2)	146.8(5.1)	0.5894
BGC3	151.3(4.3)	151.3(6.1)	1
BGC2	145.6(5.3)	151.9(4.7)	0.3199
BGC1	152.4(4.4)	145.6(6.2)	0.3633
AGC1	149.5(5.1)	147.4(4.4)	0.5136
AGC2	142.6(5.7)	140.1(4.4)	0.5245
AGC3	118.2(6.2)	129.9(3.3)	0.1093
AGC4	142.5(5.9)	146.7(5.4)	0.5136
AGC5	141.9(7)	133(5.1)	0.2293
p Value	0.0032	0.0257	–

landing in both cases (Table 3 and Fig. 2b). The HA presented significant higher IMF values for the SL landing in BGC3 (Table 4 and Fig. 2c).

3.2. Comparison of IMF values within each landing

For the RF, significant differences were concentrated in the SL (Table 2 and Fig. 2a). The Dunn's post hoc test showed differences between the intervals: BGC4 × AGC3; BGC3 × AGC3; BGC1 × AGC3; and AGC1 × AGC3. In the DL, the statistical test detected significant differences within periods; however, Dunn's test could not discriminate which conditions differed regarding the adopted significance level.

For the BF, the differences concentrated in the DL landing (Table 3 and Fig. 2b). The IMF value decreased from BGC1 to AGC3.

For HA, there were differences in both landings (Table 4 and Fig. 2c). The differences in SL occurred between the durations BGC4 × AGC3; BGC4 × AGC4, BGC3 × AGC3, BGC3 × AGC4. In DL, however, the statistical difference occurred only between the durations BGC5 and AGC3.

4. Discussion

The IMF was used to compare the RF, BF and HA recruitment strategies before and after initial GC between landings, with different mechanical constraints on male individuals with low injury

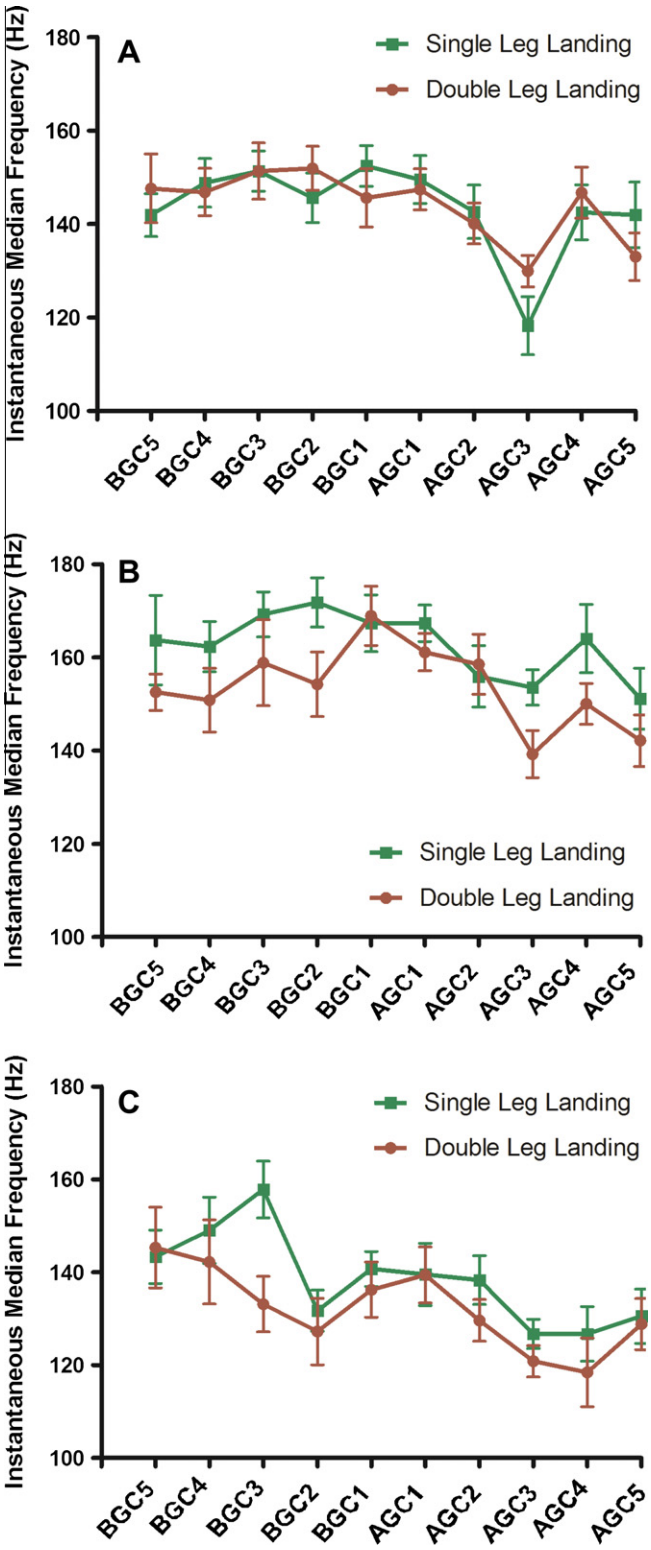

Fig. 2. Behavior of the IMF values during both landing tasks. (A) Rectus femoris, (B) biceps femoris, (C) hip adductors.

risk as evidenced by the kinematic behavior of this population, as described in Leporace et al. (2010).

The hypothesis that the muscle recruitment strategies would be different between the landings was based on the theory that the muscle synergism is indirectly controlled by local mechanoreceptors responses, which are modulated by mechanical loads imposed on the joint (Schultz et al., 1984). The IMF values were expected to be greater in the hamstrings and the HA than the RF during SL landing compared to DL. This expectation assumed that this synergism increases the dynamic joint stiffness and thus reducing the ACL strain. Higher frequencies of activation directly related to high-threshold MU (Wakeling, 2009) reflect a greater capability to generate force (Bilodeau et al., 2003). Therefore, an increase in muscle force production would result in higher frequencies of energy concentration due to an increase of the MU recruitment rate and an increase in its firing rate (De Luca, 1985; Dimitrova and Dimitrov, 2003).

Table 3
Instantaneous median frequency (IMF) of biceps femoris in each 20 ms interval. The results are expressed as mean (standard error). The *p* value described in the forth column results from a comparison of between landing tasks, while the *p* values in the last line result from within landing comparisons.

	Single leg	Double leg	*p* Value
BGC5	163.7(9.6)	152.5(3.9)	0.6092
BGC4	162.3(5.4)	150.8(6.9)	0.0637
BGC3	169.2(4.8)	158.8(9.2)	0.4264
BGC2	171.8(5.3)	154.3(6.9)	0.0256
BGC1	167.3(6.1)	168.9(6.3)	1
AGC1	167.3(3.9)	161.1(4)	0.3632
AGC2	155.9(6.6)	158.4(6.4)	0.7982
AGC3	153.5(3.8)	139.2(5.1)	0.0181
AGC4	164(7.4)	150(4.4)	0.1093
AGC5	151.1(6.6)	142.1(5.5)	0.2114
p Value	0.0899	0.0065	–

Table 4
Instantaneous median frequency (IMF) of hip adductors in each 20 ms interval. The results are expressed as mean (standard error). The *p* value described in the forth column results from a comparison of between landing tasks, while the *p* values in the last line result from within landing comparisons.

	Single leg	Double leg	*p* Value
BGC5	143.3(5.8)	145.3(8.7)	0.9658
BGC4	149(7.1)	142.2(9.1)	0.3804
BGC3	157.8(6.1)	133.1(6)	0.0096
BGC2	131.7(4.4)	127.2(7.1)	0.8984
BGC1	140.7(3.8)	136.3(5.9)	0.3266
AGC1	139.5(6.7)	139.4(6)	0.7910
AGC2	138.3(5.2)	129.6(4.5)	0.2036
AGC3	126.7(3.1)	120.8(3.4)	0.2661
AGC4	126.7(5.9)	118.4(7.4)	0.7334
AGC5	130.5(5.8)	139.4(5.5)	0.4697
p Value	0.0011	0.0154	–

The present results support this hypothesis. In general, it was observed that the central nervous system is able to modulate muscle recruitment as a function of motor activity with different mechanical constraints and joint stress. Furthermore, the strategies presented by this study on a male sample are in agreement with those suggested in the literature for females with regard to secure strategies to increase the dynamic stiffness of the knee joint and reduce strain on the ACL (Hewett et al., 2005a,b).

The IMF values of the RF showed no differences between SL and DL landings. The quadriceps muscles are primarily responsible for decreasing the center-of-gravity velocity during landings (Santello, 2005). It has been asserted that the lack of significant differences was due to the higher jump height achieved in the DL landing condition (Leporace et al., 2010). Although the literature indicates that the vertical ground reaction forces are higher during SL landings (Pappas et al., 2007), the lower height achieved during SL landing could have decreased the differences in the ground reaction forces. Therefore, it is likely that the need for an increased RF IMF value could have been attenuated.

However, the BF muscle presented different recruitment strategies between landings. In the SL task, the IMF values were higher than in the DL in either BGC2 or AGC3. These differences denote

the importance of the hamstrings in decreasing the ACL strain because anticipatory muscle activation appears to increase stiffness and reduce the risk of injury (Nigg and Liu, 1999). Likewise, increasing the IMF during AGC3 may represent a very important strategy for reducing the ligament load because evidence suggests that ACL injuries occur approximately 50 ms after GC (Krosshaug et al., 2007) and that the BF muscle plays an ACL agonist role at this interval (Solomonow et al., 1987).

The findings related to the HA support the hypothesis of "proximal stability for distal mobility" proposed by Kibler (1998). According to these authors, it is necessary to stabilize proximal joints to safely allow for greater mobility in the limbs. The present results indicate that HA act stronger before SL than DL landings, to stabilize the pelvis in anticipation of the greater subsequent activation of the hamstrings. Indeed, the HA showed higher IMF values in the SL when compared to the DL, just prior to BGC3 (40–60 ms before initial GC). These values were observed prior to the BF IMF increase, which occurred during BGC2. These results suggest a need for more pelvic stability during SL landings.

Although the RF did not present differences between landings when compared within each landing, the results were statistically different among SL periods when there was observed a reduction in the IMF during AGC3. As described above, according to Krosshaug et al. (2007), ACL injuries occur during this interval. Thus, a decrease in the IMF at this time may represent a strategy to reduce ACL strain.

For the BF, no differences were found in the comparison among the periods in the SL. However, in the DL, the IMF decreased from the interval immediately before the GC to AGC3. The synergism between BF and RF appears to be related to the effectiveness of the Central Neural System in maintaining joint stability with only minor energy cost.

The HA results confirm the importance of core-region stability in allowing appropriate lower-limb movements (Hibbs et al., 2008). During SL landings, the IMF values prior to GC were higher than after GC. Even though the IMF had presented similar behavior during DL landings, these changes were not significant. These findings suggest that anticipatory pelvic stabilization is less critical in DL than SL landings. Further studies should be conducted to investigate the IMF behavior in the other muscle groups that participate in the dynamic stability of the pelvis to validate this proposition.

The limitations of this study comprise the low sample size and the controlled laboratory environment, which do not allow a generalization of the results to even more dynamic competitive conditions. Although studies have demonstrated the effectiveness of laboratory tests in predicting injuries (Hewett et al., 2005b; Zebis et al., 2009), motor behaviors presented during sports activities are believed to differ significantly from those observed in laboratories (Lian et al., 1996). However, capturing EMG signals in competitive conditions is extremely difficult, thus justifying the use of laboratory tests.

New studies with similar designs could be conducted using other populations, such as females and athletes of different sports, to determine the influence of gender and sports in muscle recruitment strategies. In particular, females may certainly present different activation strategies in relation to male athletes. They usually present a "quadriceps-dominant" strategy of muscle recruitment, characterized by a higher quadriceps/hamstrings ratio (Landry et al., 2009), putting females under increased ACL strain and risk of injury (Renstrom et al., 2008).

5. Conclusion

The opposite differences presented by the BF and RF IMF values confirm the role of these muscles in the dynamic stability of the knee according to the mechanical overload. This study also ratified the importance of HA in stabilizing the pelvis, especially in the preground phase evidenced by the higher IMF values of the HA in the SL in relation to the DL, besides the tendency of higher IMF values before ground contact. It was concluded that muscle recruitment changes as a function of the degrees of freedom allowed by the landing tasks and that the different muscle recruitment strategies, at least in our male participants, seem to be related to protection of the ACL, which may partially explain the low incidence of lesions amongst the males as compared to females.

Conflict of interest

There are no conflicts of interest.

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